**This assignment will not be graded and is only for practice.**

**Note:** Suppose $W_1$ **and** $W_2$ are subspaces of a vector space $V$. Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$

One can verify that $W_1 + W_2$ is a subspace of $V$.

**Multiple Choice Questions (MCQ):**

1) An inner product on $\mathrm{R}^3$ is defined as: ***1 point***

$\langle ., . \rangle : \mathrm{R}^3 \times \mathrm{R}^3 \rightarrow \mathrm{R}$

$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1y_1 + x_2y_2 + x_3y_3.$

Match a set $S$ in Column A with the set $T$ given in column B, so that $T \subset Span(S)^{\perp}$. Match a set in column B with the property it satisfies in column C.

| | Set $(S)$<br><br>(Column A) | | set $(T)$<br><br>(Column B) | | Properties<br><br>(Column C) |
|---|---|---|---|---|---|
| a) | {(-2,1,3)} | i) | {(0,1,1),(2,-1,1)} | 1) | forms an orthonormal basis of a vector space which is isomorphic to $\mathbb{R}^2$ |
| b) | {(1,-2,1)} | ii) | {(1,-1,1)(2,1,1)} | 2) | orthogonal but not orthonormal |
| c) | {(1,1,-1)} | iii) | $\{\frac{1}{\sqrt{2}}$ (0,-1,1) $\}$ | 3) | not orthogonal |
| d) | {(0,1,1)(1,1,1)} | iv) | $\{\frac{1}{\sqrt{3}}(1,1,1), \frac{1}{\sqrt{2}}(-1,0,1)\}$ | 4) | orthonormal but does not form a basis of a vector space which is isomorphic to $\mathbb{R}^2$ |

Table : M2W9P1

Which of the following option are true?

- ☐ d → ii → 3
- ☐ d → iii → 4
- ☐ c → ii → 4
- ☐

a → iv → 1

☐ a → ii → 3

☐ b → iv → 2

No, the answer is incorrect.

Score: 0

Accepted Answers:

d → iii → 4
a → ii → 3

2) Suppose $W_1$ and $W_2$ are subspaces of a vector space $V$. Let $P_{W_1}$ and $P_{W_2}$ denote the projection from $V$ to $W_1$ and $V$ to $W_2$ respectively and consider the following statements:
**Statement P**: If $P_{W_1} + P_{W_2}$ is a projection from $V$ to $W_1 + W_2$, then $P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$.

**Statement Q**: $P_{W_1}^2 := P_{W_1} \circ P_{W_1}$ is not a projection from $V$ to $W_1$.

**Statement R:** If $A$ is the matrix representation of $P_{W_2}$, then $A$ can not be a symmetric matrix.

**Statement S**: $P_{W_1} - P_{W_2}$ is a projection from $V$ to $W_1 + W_2$.

Find the number of correct statements

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

***1 point***

3) Which of the followings option(s) is (are) true? ***1 point***

☐ Let $u, v,$ and $w$ be three vectors in $\mathrm{R}^2$ and $\langle ., . \rangle$ be an inner product on $\mathrm{R}^2$. If $\langle u, v \rangle = \langle u, w \rangle$, then $v = w$

☐ Let $V$ be an inner product space. If $\langle u, v \rangle = 0$ for all $u \in V$, then $v = 0$

- [ ] Let $V$ be an inner product space and $u, v \in V$. If $\langle u+v, u-v \rangle = 0$, then $\|u\| = \|v\|$
- [ ] Let $V$ be an inner product space. If $S \subset V$ then $S^{\perp} = (span(S))^{\perp}$
- [ ] Suppose $\{u_1, u_2, \ldots u_m\}$ be an orthogonal set of vectors in an inner product space $V$, then $\|u_1 + u_2 + \ldots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \ldots + \|u_m\|^2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let $V$ be an inner product space. If $\langle u, v \rangle = 0$ for all $u \in V$, then $v = 0$
Let $V$ be an inner product space. If $S \subset V$ then $S^{\perp} = (span(S))^{\perp}$
Suppose $\{u_1, u_2, \ldots u_m\}$ be an orthogonal set of vectors in an inner product space $V$, then $\|u_1 + u_2 + \ldots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \ldots + \|u_m\|^2$.

4) Which of the following option(s) is/are true? ***1 point***

- [ ] Let $A$ and $B$ be orthogonal matrices of order 3. Then $A$ and $B$ are equivalent matrices.
- [ ] Let $A$ and $B$ be orthogonal matrices. Then $A$ and $B$ are similar matrices.
- [ ] Let $A$ be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle $\theta$. Then nullity of the matrix $A$ is 0.
- [ ] Let $A$ be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle $\theta$ and $B$ be a rotation matrix corresponding to the anti- clock wise rotation of the YZ-plane about the X -axis with angle $\beta$. Then $A$ and $B$ are equivalent matrices.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let $A$ and $B$ be orthogonal matrices of order 3. Then $A$ and $B$ are equivalent matrices.
Let $A$ be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle $\theta$. Then nullity of the matrix $A$ is 0.
Let $A$ be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle $\theta$ and $B$ be a rotation matrix corresponding to the anti- clock wise

rotation of the YZ-plane about the X -axis with angle $\beta$. Then $A$ and $B$ are equivalent matrices.

**Numerical Answer Type:**

5) Let $u = (1, 2, 1)$ and $v = (1, 2, 3)$ be the vectors of the inner product space $\mathrm{R}^3$ with usual inner product. Let $T : \mathrm{R}^3 \rightarrow \mathrm{R}^3$ be an orthogonal linear transformation. If $\theta$ is an angle between $T(u)$ and $T(v)$, then find the value of $||T(u)||||T(v)|| \cos\theta$.

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

***1 point***

6) Consider a set $W = \{(1, 1, 1)\}$ from the inner product space $\mathrm{R}^3$. Find the dimension of the subspace $W^{\perp}$, where $W^{\perp}$ is the collection of the vectors which are orthogonal to the vector (1,1,1).

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

***1 point***

7) Let $v = (3, 1, 2)$ be a vector in $\mathrm{R}^3$. If $(a, b, c)$ is the vector obtained from $v$ after the anti-clock wise rotation of ZX-plane with angle $45^\circ$ about the Y- axis, then find the value of $\sqrt{2}(a + b + c - 1)$.

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

***1 point***

**Comprehension Type Question:**

With a particular frame of reference (in $\mathrm{R}^2$) the position of Rahul's house is at the point (8,1). There are two roads Road 1 and Road 2 along the lines $y = 2x$ and $y = -2x$ respectively on the $XY-$ plane.

Answer questions 8,9 and 10 using the given information.

8) Which of the following option(s) is (are) true? ***1 point***

- [ ] If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).
- [ ] If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at $\frac{6}{5}(1, -2)$.
- [ ] If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1, -2)$.
- [ ] If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at (2,4).

No, the answer is incorrect.

Score: 0

Accepted Answers:

If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).
If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1, -2)$.

9) Let $\theta$ be the acute angle between Road 1 and Road 2, then what will be the value of $5\cos(\theta)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

***1 point***

10) Let $\langle ., . \rangle$ denote the standard inner product on $\mathrm{R}^2$, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_1 + x_2y_2$. Suppose $W_1 = \{(x, y) \mid y = 2x\}$ and $W_2 = \{(x, y) \mid y = -2x\}$ are two subspaces of $\mathrm{R}^2$. Which of the following options is correct? ***1 point***

- [ ] $W_1^{\perp} = \{(x, y) \mid 2y = x\}$

- ☐ $W_2^{\perp} = \{(x, y) \mid 2y = -x\}$
- ☐ $W_1^{\perp} = \{(x, y) \mid 2y = -x\}$
- ☐ $W_2^{\perp} = \{(x, y) \mid 2y = x\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$W_1^{\perp} = \{(x, y) \mid 2y = -x\}$
$W_2^{\perp} = \{(x, y) \mid 2y = x\}$

Your score is: 0/10